Sho's Algorithm I - QFT & QPE

Road map: (1) Qubits & Quantum states.

$$|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \qquad |1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

$$H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \qquad X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

H ↳ Hadamard gate

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle.$$

$\langle 0|\psi\rangle = \alpha$ ⇓ Prob of measuring 0

$\mathbb{P}(\text{meas } 0) = \|\alpha\|^2$

$\langle 1|\psi\rangle = \beta$ ⇓ Prob of measuring 1

$\mathbb{P}(\text{meas } 1) = \|\beta\|^2.$

So: $\|\alpha\|^2 + \|\beta\|^2 = 1$

What are the implications?

Implication 1.

* Operations on quantum states need to preserve norm to 1.

(unitary operations do that).

↳ norm preserving $\quad U^\dagger = U^{-1}$

$U^\dagger U = U U^\dagger = I.$

$$|\det(U)| = 1.$$

Implication 2.

* For a single qubit, unitary gates are rotations on the surface of the Bloch sphere.

Implication 3.

* Eigenvalues and Eigenvectors of unitary matrices are special:

(a) $U|x\rangle = \underbrace{\lambda_x}_{\text{eigenvalue}}|x\rangle$ ← eigenvector

has to be of the form $\underbrace{e^{i\theta}}_{\text{looks like a phase. with magnitude 1.}}$ complex number.

(b) λ_x and λ_y. $\lambda_x \neq \lambda_y$. } if $\lambda_x \neq \lambda_y$ then corresponding eigenvectors $|x\rangle, |y\rangle$ are orthogonal.

$\langle x | y \rangle = 0$

↳ inner product.

Second Lecture.

1. Deutsch - Jozsa algorithm (function is constant or balanced).

we can do this in 1 step.

2. Grover's algorithm (unstructured search).

[1 4 9 5 8]

no pattern to take advantage of.

we can do this in $O(\sqrt{N})$ complexity.

Today: Preliminaries for Shor's Algorithm.

What we try to solve?

→ given a periodic function, finds its period.

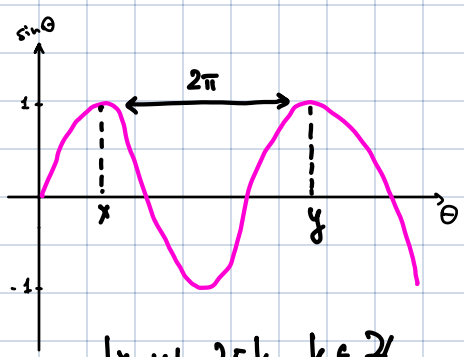
↓ $f(x)$

Strict definition:

$$f(x) = f(y) \quad \text{for } x \neq y \quad \text{iff} \quad |x-y| = kp$$

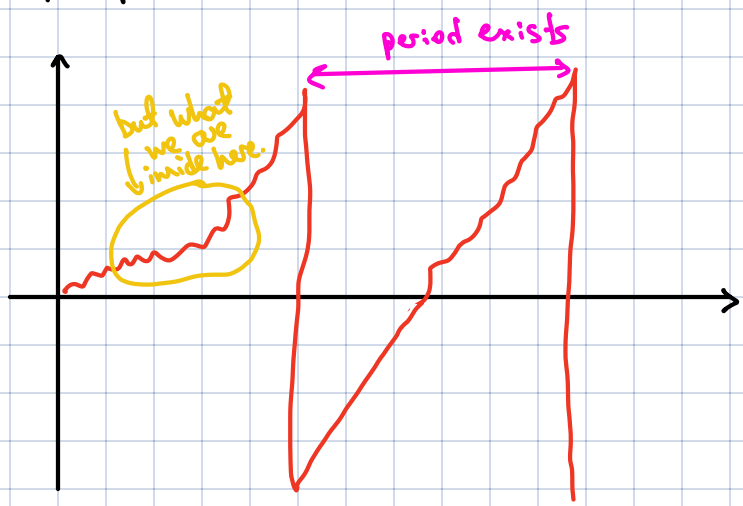
period.

example 1.



$|x-y| = 2\pi k, \; k \in \mathbb{Z}_+$

example 2.



classically: $O(\exp c \cdot n^{1/3} (\log n)^{2/3})$

quantum: Shor's algorithm $O(n^2 \log n \log\log n)$

no exponential here!

little faster than $O(n^3)$.
polynomial complexity.

Reason why this works:

(1) Quantum Fourier Transform.

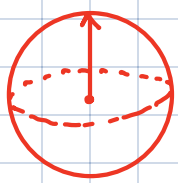
(2) Modular Exponentiation.

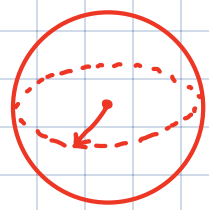
The problem of factoring a number which is a product of two prime numbers. is the basis for the security on our computers.

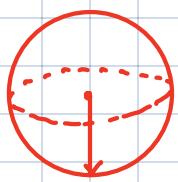
What is QFT?

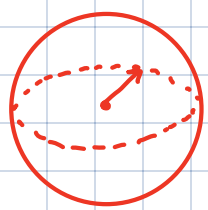
effectively a change of basis from the computational basis to the Fourier basis.

example 1-qubit: computational basis are: $\{|0\rangle, |1\rangle\}$.
Fourier basis for one qubit: $\{|+\rangle, |-\rangle\}$.

$|0\rangle =$ 

$|\tilde{0}\rangle =$ 

$|1\rangle =$ 

$|\tilde{1}\rangle =$ 

1-qubit QFT that does this?
Hadamard gate!

[play with the equatorial plane]